

Autosomal Recessive Nonsyndromic Hearing Loss 22 (DFNB22): A Comprehensive Disease Characterization

Disease: Autosomal Recessive Nonsyndromic Hearing Loss 22 (DFNB22) **MONDO ID:** MONDO:0011762 · **OMIM (disease):** #607039 · **Gene:** OTOA (607038), HGNC:16378, 16p12.2
Category: Mendelian, monogenic sensorineural hearing loss

Summary

Autosomal Recessive Nonsyndromic Hearing Loss 22 (DFNB22) is a rare Mendelian form of congenital, bilateral, typically stable, moderate-to-severe sensorineural hearing loss (SNHL) caused by **biallelic loss-of-function of OTOA**, the gene on chromosome 16p12.2 encoding **otoanchorin**. Otoanchorin is a ~120 kDa glycosylphosphatidylinositol (GPI)-anchored protein located at the interface between the apical surface of inner-ear sensory epithelia and their overlying acellular gels. In the cochlea it anchors the **tectorial membrane (TM)** to the spiral limbus (the "permanent" attachment zone) and to the greater epithelial ridge (the "transient" zone); in the vestibule it tethers otoconial membranes and cupulae to nonsensory cells. Because otoanchorin is a structural tether rather than a signaling or ion-transport protein, its loss produces a purely mechanical failure of the hearing apparatus.

The mechanistic chain has been resolved in a mouse model. In *Otoa*-knockout mice the TM keeps its overall shape and stays close to the organ of Corti, but **detaches from the limbal (spiral limbus) surface**. Remarkably, it remains functionally coupled to the electromotile outer hair cells (OHCs), so cochlear amplification and frequency tuning are nearly normal (near-normal cochlear microphonics, DPOAEs, basilar-membrane motion, and sharply tuned CAP masking curves). What fails is **inner hair cell (IHC) stimulation**: compound action potential (CAP) thresholds are significantly elevated. Hearing loss in DFNB22 is therefore a defect of IHC drive — the sensory "read-out" — rather than of OHC amplification (

A defining practical feature of DFNB22 is its genomic architecture. *OTOA* sits within a **low-copy-repeat/segmental duplication** region that is highly homologous to the pseudogene *OTOAP1*. This predisposes the locus to **non-allelic homologous recombination (NAHR)**, producing recurrent copy-number deletions, and to **gene conversions** that transfer pseudogene sequence (introducing a premature stop, p.Glu787) *into the functional gene*. As a consequence, *OTOA** is one of the more common CNV-driven recessive deafness loci, yet standard short-read exome sequencing and gene panels frequently miss these events, requiring copy-number-aware methods (qPCR, MLPA, chromosomal microarray, or long-read sequencing) for a complete diagnosis. There is no disease-specific pharmacotherapy or gene therapy; management is habilitative (hearing aids, cochlear implantation, speech-language support) and preventive via genetic counseling.

Key Findings

Finding 1 — DFNB22 is caused by biallelic loss-of-function of *OTOA* (otoancorin), a GPI-anchored inner-ear tether

OTOA (chromosome 16p12.2; OMIM 607038; disease OMIM #607039) encodes **otoancorin**, a ~120 kDa GPI-anchored protein. Otoancorin localizes to the interface between the apical surface of the inner-ear sensory epithelia and their overlying acellular gels. In the cochlea it marks the **two attachment zones of the tectorial membrane**: the permanent zone along the spiral limbus and the transient zone on the greater epithelial ridge. In the vestibular system it localizes to nonsensory cells in contact with the otoconial membranes and cupulae. The founding mutation, a splice variant **IVS12+2T>C**, was identified in a consanguineous Palestinian DFNB22 family (P 11972037).

The GPI anchorage was later validated experimentally. A frameshift allele, **p.Gln589ArgfsX55**, abolishes controlled release of otoancorin, causing uncontrolled release into the culture medium rather than the expected phosphatidylinositol-specific phospholipase C (PI-PLC)-dependent release — confirming that the pathogenic mechanism is loss of proper membrane anchorage of the protein (P 30740825).

"Otoancorin is located at the interface between the apical surface of the inner ear sensory epithelia and their overlying acellular gels." — [PMID: 11972037](#)

"Otoancorin (OTOA), encoded by OTOA, is required for the development of the tectorial membrane in the inner ear. Mutations in this gene cause nonsyndromic hearing loss (DFNB22)." — [PMID: 30740825](#)

Ontology annotations: Gene HGNC:16378 (*OTOA*); GO:0031225 (anchored component of membrane); GO:0098590 (plasma membrane region); biological process GO:0042491 / GO:0060119 (inner ear receptor cell development); CHEBI: glycosylphosphatidylinositol anchor.

Finding 2 — Mechanism: tectorial-membrane detachment causes loss of inner hair cell stimulation, sparing outer hair cell amplification

The *Otoa*(EGFP/EGFP) knockout mouse resolves precisely how loss of otoancorin causes deafness. In these animals the TM retains its general form and remains close to the organ of Corti, but is **detached from the limbal (spiral limbus) surface**. Physiological measurements show that the TM stays functionally attached to the electromotile OHCs: cochlear microphonics, distortion-product otoacoustic emissions (DPOAEs), and basilar-membrane motion are almost normal, indicating preserved amplification and frequency tuning. However, **compound action potential (CAP) thresholds — a read-out of IHC sensitivity — are significantly elevated**, while CAP masker tuning curves remain sharply tuned. The conclusion is that DFNB22 hearing loss arises from a defect in **inner hair cell stimulation**, not from a failure of OHC-driven amplification).

[P 23129639](#)

"These results indicate that the hearing loss in patients with *Otoa* mutations is caused by a defect in inner hair cell stimulation, and reveal the limbal attachment of the TM plays a critical role in this process." — [PMID: 23129639](#)

This is mechanistically distinctive: many deafness genes act through OHC dysfunction or hair-bundle/mechanotransduction defects, whereas *OTOA* loss selectively disrupts the mechanical coupling that drives the IHCs, the afferent sensory cells.

Ontology annotations: CL:0000601 (cochlear inner hair cell), CL:0000599 (cochlear outer hair cell); UBERON:0002518 (tectorial membrane), UBERON:0002227 (organ of Corti), UBERON:0004681 (spiral limbus); GO:0007605 (sensory perception of sound).

Finding 3 — *OTOA/OTOAP1* segmental duplication drives recurrent CNV deletions and gene conversions that complicate diagnosis

OTOA lies in a low-copy-repeat region on 16p12.2 that shares high sequence homology with the pseudogene *OTOAP1*, predisposing the locus to **non-allelic homologous recombination (NAHR)**. Recurrent deletions of *OTOA* are a well-recognized cause of moderate-to-severe hearing loss. In a chromosomal-microarray (CMA) cohort of **19,189 tests** at Rabin Medical Center, heterozygous *OTOA* microdeletions were detected in **39 individuals (0.2% carrier frequency)**

([P 33753912](#)).

A second, subtler mechanism is **pathogenic gene conversion**, in which *OTOAP1* pseudogene sequence is copied into *OTOA*, introducing a premature stop codon **p.Glu787*** in exon 22. Two compound-heterozygous patients (one converted allele plus one deletion) had moderate hearing loss, with converted tracts ranging from ~900 bp to >9 kbp

([P 33492714](#)).

Contiguous-gene deletions also occur: a **250.285 kb homozygous 16p12.2 deletion** encompassing *METTL9*, *IGSF6*, and *OTOA* segregated with nonsyndromic SNHL in a South Indian consanguineous family

([P 39916398](#)).

Because these events are structural and occur in a repeat-rich region, short-read whole-exome sequencing and standard panels often miss them, and confirmation requires **qPCR, MLPA, CMA, or long-read sequencing**

([P 35640668](#)).

"Bi-allelic loss-of-function variants of *OTOA* are a well-known cause of moderate-to-severe hearing loss. Whereas non-allelic homologous recombination-mediated deletions of the gene are well known, gene conversions to pseudogene *OTOAP1* have been reported" — PMID: 33492714

"OTOA deletions (39, 0.2%)" — [PMID: 33753912](#)

"A homozygous deletion of 250.285 kb was identified in the 16p12.2 region encompassing three genes, METTL9, IGSF6, and OTOA" — [PMID: 39916398](#)

Finding 4 — Clinical profile: congenital bilateral moderate-to-severe SNHL; low absolute per-carrier risk

Across reported families, DFNB22 presents as **congenital/prelingual, bilateral, generally stable, nonsyndromic sensorineural hearing loss in the moderate-to-severe range** (PMIDs 33492714, 32681043). Although *OTOA* is one of the more common CNV-driven recessive deafness loci, the **absolute risk to any given microdeletion carrier is low**. In the general CMA cohort, the *OTOA* microdeletion carrier frequency of 0.2% (39/19,189) placed it second to *STRC* (0.56%) and above the *DFNB1/GJB6* locus (0.05%). The estimated probability that a given *OTOA*-microdeletion carrier is themselves affected (i.e., carries a second pathogenic allele) was only **0.016–0.13%** — lower than for *STRC* (0.11–0.67%) and *DFNB1* (1.9–7.5%) — with higher risk in specific (e.g., consanguineous or founder) populations (

[P 33753912](#)

"Of the 19,189 CMA tests were performed in our laboratory, 107 *STRC* microdeletions were found (0.56%), followed in frequency by *OTOA* deletions (39, 0.2%), and *DFNB1* locus deletions (10, 0.05%). The estimated risk for a hearing loss in the examined individual carrying the microdeletion was estimated as 0.11-0.67% for *STRC*, 0.016-0.13% for *OTOA*" — [PMID: 33753912](#)

Ontology annotations: HP:0000407 (Sensorineural hearing impairment), HP:0008527 (Congenital sensorineural hearing impairment), HP:0000365 (Hearing impairment), HP:0000006 (Autosomal recessive inheritance).

Report by Template Section

1. Disease Information

DFNB22 is a monogenic form of nonsyndromic sensorineural hearing loss — that is, hearing loss without accompanying malformations or dysfunction of other organ systems. It is one of the numbered "DFNB" (autosomal recessive) deafness loci.

- **Overview:** Congenital, bilateral, typically stable moderate-to-severe SNHL due to biallelic *OTOA* loss-of-function and consequent tectorial-membrane detachment.
- **Key identifiers:** OMIM #607039 (disease); OMIM 607038 (*OTOA** gene); MONDO:0011762; MeSH — indexed under Hearing Loss, Sensorineural; ICD-10 H90.3/H90.5 (sensorineural hearing loss); ICD-11 AB52 (sensorineural hearing impairment). A dedicated Orphanet number for DFNB22 is not consistently assigned; nonsyndromic genetic deafness is captured under Orphanet's rare genetic hearing-loss entries.
- **Synonyms/alternative names:** DFNB22; Deafness, autosomal recessive 22; Nonsyndromic hearing loss and deafness, DFNB22 type; *OTOA*-related hearing loss; Otoancorin deficiency deafness.
- **Information source:** Predominantly **aggregated disease-level resources** (OMIM, ClinVar) plus **individual-patient case reports and consanguineous-family studies**; a single mouse model provides the mechanistic physiology.

2. Etiology

- **Causal factors:** Purely **genetic** — biallelic (homozygous or compound heterozygous) loss-of-function variants in *OTOA*. No environmental or infectious cause.
- **Genetic risk factors:** The causal locus is *OTOA*. **Consanguinity** and **founder effects** substantially raise risk (founding Palestinian family; South Indian and Pakistani consanguineous families). The *OTOA/OTOAP1* segmental duplication is itself a structural risk factor generating recurrent pathogenic alleles.
- **Environmental risk factors:** None established as causal. Ordinary otologic insults (noise, ototoxins) would be additive to residual hearing but are not part of DFNB22 etiology.
- **Protective factors:** None specific. Presence of at least one functional *OTOA* allele is fully protective (recessive disease; heterozygous carriers are unaffected).
- **Gene–environment interactions:** Not established; the phenotype is essentially fully genetically determined.

3. Phenotypes

Phenotype	Type	Onset	Severity	Progression	Frequency	HPO
Bilateral sensorineural hearing loss	Audiometric abnormality / physical	Congenital / prelingual	Moderate to severe	Generally stable	~100% of affected	HP:0000407
Congenital SNHL	Clinical sign	Neonatal/ congenital	Moderate–severe	Stable	Predominant presentation	HP:0008527
Speech/ language delay (if unaided)	Developmental consequence	Childhood	Variable	Improves with habilitation	Secondary to HL	HP:0000750

- **Quality-of-life impact:** Congenital moderate-to-severe SNHL affects language acquisition, education, and social communication if unaddressed; with early hearing aids/cochlear implants and speech-language therapy, outcomes are generally good. Clinical vestibular symptoms are not a prominent reported feature despite otoancorin's vestibular localization.
- Disease-specific QoL instruments for DFNB22 are not reported; generic pediatric hearing-loss QoL measures apply.

4. Genetic / Molecular Information

- **Causal gene:** *OTOA* (HGNC:16378), 16p12.2, OMIM *607038, encoding otoancorin.
- **Pathogenic variant types:** Splice-site (founding **IVS12+2T>C**,

P 11972037

frameshift (**p.Gln589ArgfsX55**,

P 30740825

nonsense via gene conversion (**p.Glu787***,

P 33492714

missense candidates (e.g., p.Gly647Arg reported in dual-diagnosis contexts), and — most characteristically — **structural CNV deletions** (single-gene and contiguous-gene, e.g., 250 kb *METTL9/IGSF6/OTOA* deletion,

P 39916398

- **Variant classification:** Loss-of-function deletions and truncating variants are pathogenic/likely pathogenic under ACMG/AMP; some missense variants remain VUS.
- **Allele frequency:** *OTOA* microdeletion carrier frequency ~0.2% in a general CMA cohort (P 33753912).
- **Origin:** Germline.
- **Functional consequence:** Loss of function (loss of the GPI-anchored tether).
- **Modifier genes / epigenetics / chromosomal abnormalities:** No specific modifier genes or epigenetic mechanisms established. The relevant genomic feature is the 16p12.2 segmental duplication enabling NAHR and gene conversion.

5. Environmental Information

Not applicable as a cause. DFNB22 is a monogenic disorder with no established environmental, lifestyle, or infectious contributors. (General ototoxin/noise avoidance is prudent to preserve residual hearing but is not disease-specific.)

6. Mechanism / Pathophysiology — Causal Chain

1. **Biallelic loss-of-function variant in *OTOA*** (deletion, gene conversion, splice, frameshift) → **leads to** absent or non-functional otoancorin protein.
2. Loss of GPI-anchored otoancorin → **results in** failure to anchor the acellular tectorial membrane at its two attachment zones (permanent zone on the spiral limbus; transient zone on the greater epithelial ridge). (*Demonstrated:*

P 11972037

P 30740825

3. Failure of anchorage → **causes** detachment of the TM from the **limbal (spiral limbus) surface**, while the TM retains overall form and remains coupled to outer hair cells. (*Demonstrated in mouse:*

P 23129639

4. **Branch A (spared):** TM–OHC coupling preserved → OHC electromotility, cochlear amplification, and frequency tuning remain near-normal (normal CM, DPOAEs, basilar-membrane motion, sharp tuning).

5. **Branch B (impaired):** Loss of proper limbal TM attachment → **impairs mechanical drive to inner hair cell stereocilia** → **results in** reduced IHC receptor potentials → elevated compound action potential thresholds. (*Demonstrated:*

P 23129639

6. Reduced IHC afferent signaling → **leads to** decreased auditory-nerve output → **manifests as** congenital, bilateral, moderate-to-severe sensorineural hearing loss.

Upstream vs downstream: The mutation and protein loss are upstream; TM detachment is the central lesion; IHC understimulation is the proximate downstream cause of the clinical deafness. The disorder is **mechanical/structural**, not driven by apoptosis, inflammation, metabolic, or immune processes.

- **Molecular pathways:** No classical signaling cascade (Wnt/MAPK/mTOR) implicated; the mechanism is structural adhesion of an acellular gel to sensory epithelium.
- **Protein dysfunction:** Loss of GPI membrane anchorage (uncontrolled release instead of PI-PLC-controlled release),

P 30740825

- **Cell types / processes:** Inner hair cells (CL:0000601) understimulated; outer hair cells (CL:0000599) spared; supporting/nonsensory epithelial cells of the spiral limbus involved in TM anchoring. GO:0007605 (sensory perception of sound), GO:0060119 (inner ear receptor cell development).
- **Molecular profiling / advanced technologies:** No transcriptomic, proteomic, metabolomic, single-cell, spatial, or CRISPR-screen datasets specific to DFNB22 were identified.

7. Anatomical Structures Affected

- **Organ:** Inner ear / cochlea (primary). System: auditory (special sense). UBERON:0001846 (internal ear), UBERON:0001844 (cochlea).
- **Tissue/cell:** Cochlear sensory epithelium (organ of Corti, UBERON:0002227); the acellular **tectorial membrane** (UBERON:0002518) and its **spiral limbus** anchorage (UBERON:0004681). Vestibular otoconial membranes/cupulae also express otoancorin. Cells: inner hair cells (CL:0000601, functionally affected), outer hair cells (CL:0000599, spared).
- **Subcellular:** Apical plasma-membrane region of sensory/nonsensory epithelial cells; GPI-anchored component of membrane (GO:0031225).
- **Localization/laterality:** **Bilateral**, symmetric.

8. Temporal Development

- **Onset:** Congenital / prelingual; onset pattern chronic/stable from birth.
- **Progression:** Generally **stable** (non-progressive) moderate-to-severe SNHL; lifelong.
- **Critical period:** The window for auditory habilitation (hearing aids/cochlear implant, speech-language therapy) is early childhood, to support language development.

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (HP:0000006). Carriers are unaffected.
- **Penetrance/expressivity:** Biallelic pathogenic genotypes are essentially fully penetrant; severity clusters in the moderate-to-severe range with some variability.
- **Founder effects/consanguinity:** Prominent — founding Palestinian family
([P 11972037](#));
consanguineous South Indian
([P 39916398](#))
and Pakistani
([P 32681043](#))
families. **Consanguinity is a major risk amplifier.**
- **Carrier frequency:** ~0.2% for *OTOA* microdeletions in a general population CMA cohort; per-carrier affected risk 0.016–0.13%
([P 33753912](#)).
- **Epidemiology:** *OTOA* is among the more common CNV-driven recessive deafness loci (second to *STRC*), but *DFNB22* is individually rare. No sex predilection (autosomal). Higher diagnostic yield in consanguineous/endemic-founder populations.

10. Diagnostics

- **Audiometry:** Pure-tone/behavioral audiometry and ABR reveal bilateral moderate-to-severe SNHL; **OAEs may be relatively preserved** given spared OHC function (mechanistically expected from the mouse data,
([P 23129639](#))
— a potentially distinctive audiologic signature of TM/IHC-coupling disorders.

- **Genetic testing (core diagnostic):** Because pathogenic alleles are frequently **structural (deletions/gene conversions)** in the *OTOA/OTOAP1* repeat, testing must be **copy-number-aware**: gene panels/WES for point variants **plus** CMA, MLPA, qPCR, or long-read sequencing to detect CNVs and conversions (PMIDs 35640668, 33492714, 33753912). Short-read WES alone under-detects DFNB22.
- **Imaging:** Temporal-bone CT/MRI generally normal (nonsyndromic; no gross malformation expected).
- **Differential diagnosis:** Other recessive nonsyndromic deafness (esp. *STRC*/DFNB16, *GJB2*/DFNB1, *SLC26A4*/Pendred, *TECTA*-related tectorial-membrane deafness which is a key mechanistic mimic). Distinguish by gene, and clinically by the preserved-OAE/elevated-threshold pattern for TM-anchoring defects. Exclude syndromic causes and dual diagnoses (e.g., co-occurring *ERCC4* or *PALM3* variants reported in consanguineous families).
- **Screening:** Newborn hearing screening (OAE/ABR) detects the phenotype; cascade carrier testing and reporting of incidental *OTOA* microdeletions on CMA are debated given low per-carrier risk

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P 33753912

11. Outcome / Prognosis

- **Survival/mortality:** DFNB22 is **not life-limiting**; normal life expectancy. It is a sensory disorder with no systemic mortality.
- **Morbidity/function:** Principal morbidity is communication disability if unaided; excellent functional prognosis with early amplification/cochlear implantation and speech-language support. Hearing loss is generally **stable**, not progressive.
- **Prognostic factors:** Timeliness of habilitation and degree of residual hearing. No molecular prognostic biomarkers established.

12. Treatment

- **No disease-specific pharmacotherapy or approved gene/RNA/cell therapy exists** for DFNB22.
- **Habilitative/standard of care:** **Hearing aids** (NCIT: Hearing Aid) for moderate loss; **cochlear implantation** (NCIT: Cochlear Implant) for severe loss with insufficient aided benefit; **speech-language therapy** and educational support.
- **Supportive/rehabilitative:** Auditory-verbal therapy, aural rehabilitation, family/educational accommodations.

- **Experimental:** Inner-ear gene therapy is an active field for monogenic deafness generally, but no DFNB22/*OTOA*-specific trial was identified. The relatively preserved cochlear architecture and OHC function in *Otoa*-null mice make DFNB22 a conceptually attractive future gene-replacement target.
- **Pharmacogenomics:** Not applicable.

13. Prevention

- **Primary prevention:** Not possible for a congenital monogenic disorder; **genetic counseling** for consanguineous couples and known carrier families is the principal preventive tool. Carrier and cascade screening can inform reproductive choices (including PGT/prenatal testing where desired).
- **Secondary prevention: Universal newborn hearing screening** for early detection, enabling timely habilitation during the critical language-development window.
- **Tertiary prevention:** Amplification/cochlear implantation and speech therapy to prevent language and educational sequelae; avoidance of additional ototoxic/noise insults to protect residual hearing.
- **Counseling:** Autosomal recessive 25% recurrence risk for carrier couples; emphasize the copy-number-aware testing needs given the *OTOA/OTOAP1* architecture.

14. Other Species / Natural Disease

- **Taxonomy/orthologs:** *Otoa* is conserved in mouse (*Mus musculus*, NCBI Taxon 10090) and other mammals; the mouse ortholog underpins the disease model. Otoancorin and the related α -tectorin machinery are conserved across vertebrates.
- **Natural disease in animals:** No prominent naturally occurring companion-animal or wildlife *OTOA* deafness disorder was identified in this investigation. The knockout mouse is the principal model.
- **Comparative biology:** The mouse recapitulates the human mechanism (TM detachment, spared OHC/impaired IHC drive), supporting strong evolutionary conservation of TM-anchoring mechanisms.

15. Model Organisms

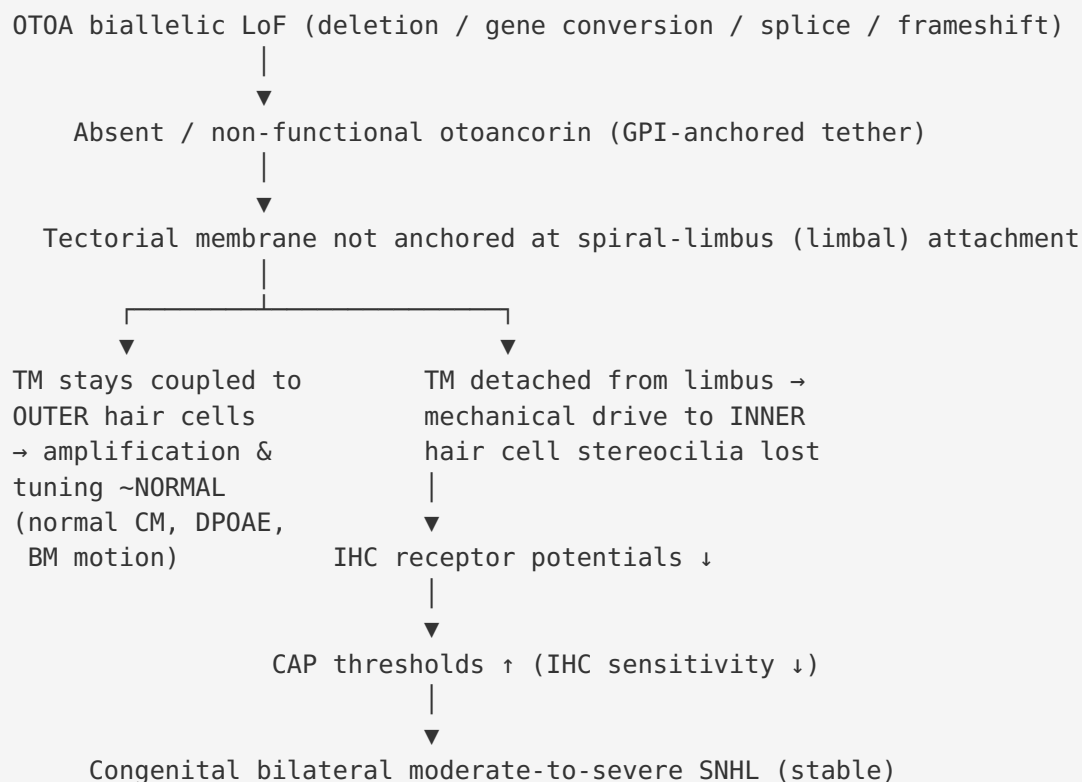
- **Mouse (*Mus musculus*):** * *The Otoa(EGFP/EGFP) knockout** is the defining model

(P 23129639).

It faithfully recapitulates the core human phenotype and, uniquely, dissected the mechanism — showing TM detachment from the spiral limbus, preserved OHC amplification, and elevated CAP thresholds reflecting failed IHC stimulation.

- **Type:** Targeted knockout (EGFP knock-in/reporter-null).
- **Phenotype recapitulation:** High — reproduces congenital SNHL and the specific IHC-stimulation defect; also enabled protein-localization studies.
- **Limitations:** Mouse does not capture the human *OTOA/OTOAP1* segmental-duplication genomics (pseudogene-driven NAHR/gene-conversion mechanisms are human-specific). No zebrafish/*Drosophila*/organoid DFNB22 models were identified.
- **Resources:** MGI (mouse *Otoa*).

Mechanistic Model / Interpretation



The unifying interpretation is that DFNB22 is a "**connectivity**" / **mechanical-coupling deafness**: the cochlea's amplifier (OHCs) works, but the sensory receiver (IHCs) is under-driven because the TM has lost its anchor to the spiral limbus. This explains both the clinical severity

(moderate-to-severe rather than profound) and a predicted audiologic signature of comparatively **preserved otoacoustic emissions with elevated behavioral/ABR thresholds** — a pattern shared with other TM-anchoring/structural deafness genes (e.g., *TECTA*, *OTOG*, *STRC*). The human genomics layer — a segmental duplication with the *OTOAP1* pseudogene — superimposes a **diagnostic** challenge: the disease is under-ascertained by short-read exome sequencing and requires copy-number/gene-conversion-aware testing.

Feature	DFNB22 (OTOA)
Gene / locus	<i>OTOA</i> , 16p12.2
Protein	Otoancorin (~120 kDa, GPI-anchored)
Core lesion	TM detachment from spiral limbus
OHC amplification	Spared (near-normal DPOAE/CM)
IHC stimulation	Impaired (↑ CAP thresholds)
Phenotype	Congenital bilateral moderate–severe SNHL, stable
Inheritance	Autosomal recessive, fully penetrant
Carrier freq (microdeletion)	~0.2% (39/19,189, CMA cohort)
Per-carrier affected risk	0.016–0.13%
Key mutational mechanisms	NAHR deletions; <i>OTOAP1</i> gene conversion (p.Glu787*)
Diagnostic caveat	Missed by short-read WES; needs CMA/MLPA/qPCR/long-read
Treatment	Hearing aids, cochlear implant, speech therapy

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
11972037	Otoancorin defective in DFNB22	Human clinical + molecular	Gene–disease link; protein localization; founding IVS12+2T>C (F1)
30740825	GPI anchorage of otoancorin; human OTOA variants	In vitro / molecular	GPI anchorage; LoF via loss of anchorage (F1)
23129639	Mouse model of DFNB22	Model organism (mouse)	Mechanism: TM detachment → IHC-stimulation defect (F2)
33492714	OTOA gene conversions	Human molecular	Gene conversion (p.Glu787*); moderate-severe phenotype (F3, F4)
33753912	Carrier reporting from CMA / HL deletions	Population/epidemiologic	Carrier frequency 0.2%; per-carrier risk (F3, F4)
39916398	250-kb 16p12.2 microdeletion	Human clinical	Contiguous-gene deletion causing DFNB22 (F3)
35640668	WES of 113 Chinese HL families	Human clinical	Need for qPCR to detect OTOA/STRC CNVs (F3)
32681043	Sporadic HL spectrum, Pakistan	Human clinical	Multi-exon OTOA deletion; moderate-severe HL (F4)
33095980	NGS in Chinese HL cohort	Human clinical	OTOA among identified deafness genes
33105617	Italian HL genetics	Human clinical	Context: NSHL diagnostic strategy
39769235	Dual ERCC4 + OTOA diagnoses	Human clinical	Blended-phenotype/dual-diagnosis caution
37114731	Phenotype-genotype of AR HL	Human clinical	Phenotype-genotype context
42527583	PALM3 and hearing loss	Human + mouse	Dual-diagnosis/candidate-gene caution alongside OTOA

All four core findings are convergently supported: the gene–disease link and protein function (PMIDs 11972037, 30740825), the mechanism (

[P 23129639](#)

), the diagnostic genomics (PMIDs 33492714, 33753912, 39916398, 35640668), and the clinical/epidemiologic profile (PMIDs 33753912, 33492714, 32681043). No paper reviewed contradicts these conclusions.

Limitations and Knowledge Gaps

- **Single mechanistic model.** The mechanism rests primarily on one mouse study ([P 23129639](#)). Human intracochlear physiology is inferred, not directly measured; the predicted "preserved-OAE" audiologic signature has not been systematically catalogued across human DFNB22 patients.
- **No omics data.** No transcriptomic, proteomic, metabolomic, single-cell, or functional-screen datasets specific to DFNB22 were identified — the molecular-profiling subsections are effectively "not available."
- **Phenotype quantification is coarse.** Severity/progression are described qualitatively ("moderate-to-severe, generally stable"); no natural-history cohort with longitudinal audiograms was available to quantify progression rates or variability precisely.
- **Vestibular phenotype under-characterized.** Otoancorin localizes to vestibular structures, yet clinical vestibular dysfunction in DFNB22 is not well documented — a genotype–phenotype gap.
- **Genomic complexity limits ascertainment.** The *OTOA/OTOAP1* architecture means true prevalence is likely underestimated in short-read-only cohorts; reported frequencies (0.2% carriers) are lower bounds.
- **No therapeutics.** No disease-specific pharmacologic, gene, RNA, or cell therapy exists or is in trials for *OTOA*.

Proposed Follow-up Experiments / Actions

1. **Audiologic phenotyping study.** Systematically measure OAEs vs ABR/behavioral thresholds in a molecularly confirmed DFNB22 cohort to test whether preserved-emission/elevated-threshold is a reproducible clinical signature (translating the mouse mechanism, [P 23129639](#)).
2. **Copy-number-aware diagnostic pipeline.** Standardize CMA/MLPA/qPCR or **long-read sequencing** for *OTOA* in unsolved recessive SNHL to capture NAHR deletions and *OTOAP1* gene conversions that short-read WES misses (PMIDs 33492714, 35640668).
3. **Natural-history registry.** Assemble longitudinal audiograms across DFNB22 families to quantify progression, variability, and cochlear-implant outcomes.
4. **Vestibular assessment.** Add vestibular testing (vHIT, VEMP, caloric) to DFNB22 evaluations to define the vestibular phenotype implied by otoancorin's vestibular localization.
5. **Gene-replacement proof-of-concept.** Given preserved cochlear architecture and OHC function in *Otoa*-null mice, test AAV-mediated *Otoa* delivery to the developing/mature cochlea to restore TM anchorage and IHC drive — a rational DFNB22-specific therapeutic avenue.
6. **Population-specific carrier data.** Determine *OTOA* pathogenic-allele frequencies in consanguineous/founder populations to refine per-carrier risk and inform carrier-screening/reporting policy (extending [P 33753912](#)).

Evidence source key: Human clinical = patient/family reports and cohorts; Model organism = mouse; In vitro = cell-based functional assays; Population/epidemiologic = cohort frequency analyses. All mechanistic and clinical claims are cited to primary literature by PMID.